

Emilien Breton

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— EXPERIENCE —

- EcoSafeSense**
FIRMWARE ENGINEER

Ottawa | October 2024–Present

 - Writing firmware for an ESP32-based air quality sensor and complementary test bench.
- Cohere**
DATA QUALITY SPECIALIST — ADVANCED MATHEMATICS

Freelance | October 2024–April 2025

 - Wrote, audited and corrected LLM prompts and responses to produce **spotless training data** in formal logic, combinatorics, number theory, graph theory and mathematical optimization.

— PROJECTS —

- DFA Regex Engine**
 - Built a regex engine in C99 that compiles regular expressions down to minimal deterministic finite automata to match input strings in **linear time** without backtracking.
 - Wrote a test suite of over **500 end-to-end tests** to ensure correctness of the engine and catch regressions.
 - Developed a grep-like tool to stress-test the engine and achieved performance on par with GNU grep.
- Multilayer Perceptron**
 - Wrote a static reverse-mode automatic differentiation library in C99 for use in deep learning.
 - Designed a multilayer perceptron model atop the library and implemented stochastic gradient descent with momentum, ultimately achieving **97% accuracy** on the MNIST database after 3 minutes training on 16 threads.
- Purely Functional Language**
 - Developed an interpreter for a lazily-evaluated λ -calculus augmented with monadic I/O.
 - Bootstrapped the interpreter into a usable language by writing a **2000-line prelude** complete with booleans, integers, pairs, optionals, lists, strings, functors, monads, monoids, foldables, traversables...
- Breadboard Microcomputer**
 - Designed an 8-bit microcomputer from logic gates upwards, including an assembler, emulator and rudimentary C compiler in Rust, totaling over **20 000 SLOC** and **750 hours** of work.
 - Wrote various programs running natively on the microcomputer, including a Wozmon-like memory monitor, 16×16 sprite editor, native assembler, postfix notation calculator and Tetris clone.
 - Built the microcomputer in hardware by hand-wiring discrete 74HC-series logic chips on breadboards.

— VOLUNTEERING —

- Computer Science Club**
CLUB EXECUTIVE

University of Ottawa | June 2022–Present

 - Running a growing community of **over 1500** computer science students at the University of Ottawa.
 - Collaborating with the executive board to brainstorm, plan, fund and market monthly events and meetups, such as workshops on Vim bindings and a mini-course on the λ -calculus.
- Hack the Hill Hackathon**
DEVELOPMENT MANAGER — DEVELOPMENT TEAM

Ottawa | November 2022–October 2024

 - Led development of an open-source event management system based on Next.js and Prisma and used by over **1000 hackers** and **50 organizers** throughout the hackathon.
 - Built and maintained an internal payment portal powered by Stripe and React.js in collaboration with the sponsorship team that processed upwards of **20 000\$**.

— AWARDS —

- **IEEE x HTB CTF** — 2nd place
 - **Tech-NoI-Hack 2025** — 2nd place
 - **AIT Hackathon 3.0** — 1st place, Mocha challenge
 - **uOCTF 2025** — 1st place
 - **uOttaHack 6** — 1st place, QNX challenge
 - **CS Games 2024** — 1st place, IoT challenge
 - **uOCTF 2024** — 1st place
 - **DeFi The Conventional** — 1st place, DeFi challenge
- January 2026

October 2025

September 2025

March 2025

January 2025

March 2024

November 2023

March 2022

— SKILLS —

- **Languages** — C • Rust • Python • Haskell • JavaScript
 - **Tools** — GNU/Linux • GDB • GNU Make • Vim • React
- 613463ca • Feb 2026